Deep cumulants under a fixed FLOP budget

A quantitative assessment of representations, leaderboard hypotheses, and the next experiment

Assessment. I do not have a credible representation that preserves the required quenched third cumulant while eliminating the 107-unit old tier. The product spectra rule out one compression route, not every contraction-specific representation. The strongest remaining checks concern conditional variance and exact algebraic reuse.

Derived. V29 scores 5.41×10^{-9}; the quoted leader ranges imply scores of $2.13–3.06\times10^{-9}$. Even free old-source transport leaves V29 at 3.18×10^{-9} at 153 units. Thus cost reduction alone does not explain the best result. A 2% D21 perturbation adds about 1.68×10^{-9} raw MSE under your measured law.

Costs below use $U = 2^{31}$ FLOPs, float32, classical matmul, and rounded leading terms; float64 doubles them. These are derivations and proposed falsifiers, not measurements on your network. No checkpoints or implementation were supplied.

Q1(a). Moving the nonlinearity into the basis

Derived. With $T = Q^{\otimes 3}K$, the cubic Hermite correction to neuron i evaluates $K(q_i,q_i,q_i)$. Moving coordinates does not remove mixed cubic coefficients. For $G = W\,\mathrm{diag}(w_1)$, exact retained-subspace transport is

$$GQ = Q^{+}R, \qquad K^{+} = R^{\otimes 3}K + K_{\text{birth}}.$$

A generic symmetric K has r(r+1)(r+2)/6 coefficients. Straight dense core rotation costs $3r^4/n^3$ units; quadratic-vector queries cost r^3/n^2, followed by r/n to expand D21. At r=224 these are 7.03, 10.72, and 0.219 units *per layer*, before basis fitting and births; at r=384 the first two alone cost 114.75. Hermite truncation loses the mixed cubic information already implicated by your rank cliffs.

First falsifier: your existing rank-cliff experiment already rejects using r≈100 merely because it retains 99% of leg energy. No rerun is needed (0 additional units). That energy criterion is not a 2% contraction-error guarantee. I am not proposing another Tucker trial.

Q1(d). An exact streamed identity, with limited savings

Let q_i be row i of Q, and w the birth coefficient vector. Compute one row or row tile at a time:

$$a = F_A^{I} q_i, \quad p = F_P^{I} q_i, \quad z = F_A(w \circ a \circ p), \quad H_{i,:} = (Qz)^{I}.$$

This computes $H=(A\circ P\circ w)A^{\mathsf{T}}$ exactly without allocating dense A or P. Including two elementwise products, its cost is $[8n^2r-n^2-nr]/U$: approximately 0.875 units at r=224 or 1.50 at r=384, *per source*. Transport still requires GQ and factor rotations $RF_{A,P}$; the latter add $2(r/n)^2$ per source, plus shared basis work.

Loss: none beyond the existing basis truncation. **First falsifier:** compare against one cached dense H, under 3 units; reject if the float32 relative difference exceeds 10^{-5}. This saves memory, not the old-tier asymptotics. I know no general $O(nr^2)$ identity here without extra core structure; this is not an arithmetic lower-bound claim.

Q1(b). What self-averaging actually removes

Derived. Freeze a checkpoint and its gates. Absorb the gates and the fresh-row covariance square root into a symmetric tensor T, so two distinct new weight rows become independent g,h~N(0,I_r). Write y=T(g,g,h), and $\tau_c=\Sigma_a T_{aac}$. Gaussian pairings give [1]:

$$\mathbb{E}_g[y \mid h] = \tau^I h, \qquad \mathbb{E}[y] = 0,$$

$$\mathbb{E}(y - \tau^T h)^2 = 2\|T\|_F^2, \qquad \mathbb{E}y^2 = 2\|T\|_F^2 + \|\tau\|^2.$$

The first identity contracts $E[g_a g_b]=\delta_{ab}$; the two remaining fourth-moment pairings produce the variance. For symmetric T, $\|\tau\|^2 \le (r+2)\|T\|^2/3$; the trace-adjoint norm attains equality at T=sym(I⊗v). Consequently the RMS error of this conditional-expectation replacement obeys

$$\rho = \sqrt{\frac{2\|T\|_F^2}{2\|T\|_F^2 + \|\tau\|^2}} \;\ge\; \sqrt{\frac{6}{r+8}}.$$

The floor is 7.62% at r=1024, 16.08% at r=224, and 23.57% at r=100. Here r is the retained support dimension. It also applies to the *sum* of old tensors before the fresh layer. Averaging more matrix entries stabilizes the measured energy; it does not erase signed quenched error. If old content contributes fraction α of total D21 RMS, even the optimistic r=224 floor requires α≤0.124 to reach 2% overall.

This only rejects averaging away g. A predictor retaining both realized rows and extra tensor information can do better. Likewise, the theorem concerns one fresh linear layer with gates frozen; it does not assume independence after future gates have adapted.

State/update/cost. The tempting state is τ, with a trace-only readout costing under 0.002 units/layer. But it is not closed. The contraction map has components $\Sigma_{ab} T_{abc} B_{ab}$. Exact transport requires

$$\tau^+ = G\,\mathcal{T}(G^T G),$$

not merely Gτ. Replacing G^TG by its expectation loses precisely a quadratic-query component. A covariance/Gram-conditioned predictor must specify which realized information it retains. Gaussian scale mixtures and memoryless slice regeneration remain rejected by your measurements. **First falsifier:** the ≤30-unit conditional-resampling audit in Q3.

Q1(c). Earlier slices are queries, not a closed state

Derived. Exact old-content transport is $T^+=G^{\otimes 3}T$, hence $D21^+_{ic}=T(g_i,g_i,g_c)$. Covariance and current slices cannot determine it: a tensor supported on permutations of three distinct coordinates has zero D3/D21, yet acquires nonzero slices after generic mixing. Testing that witness costs under 0.01 units; it disproves universal closure, not an empirical 2% guarantee.

In an r-dimensional support, each earlier D21 bank supplies at most nr independent numbers. Generic exact recovery therefore needs at least ceil[(r+1)(r+2)/(6n)] banks: 2 at r=100, 9 at r=224, 25 at r=384. These are necessary dimension counts, not sufficiency results. Reconstructing K returns to Q1(a)'s cost.

For fixed future maps, the exact minimal *linear* state is the rank of the stacked future-query operator. Its useful 2% dimension depends on tensor energy in those observable directions; the supplied spectra do not determine it. Precomputing future slices merely pays their contractions earlier, and adaptive gates complicate even that construction.

Q1(d), continued. A conditional fusion worth checking

Derived, conditional. Inspect whether the code has M=A diag(a)+P diag(b)+R_0, with $R_0=UV^T$ of rank t≤16, and source D21=2H+T2+(T3+2T4)/3. If so, distributivity gives

$$D21 = XA^T + YP^T + \tfrac{1}{3}(P \circ P)R_0^T,$$

$$X = 2w \circ A \circ P + \tfrac{1}{3}a \circ P \circ P,$$

$$Y = w \circ A \circ A + \tfrac{2}{3}a \circ A \circ P + b \circ P \circ P + \tfrac{2}{3}R_0 \circ P.$$

Vectors broadcast over columns. With dense legs already available, two dense products plus the thin residual cost under 2.07 units/source, versus four dense products before fusion. For old legs, reconstructing two legs, projecting X and Y, and expanding their *sum* costs about 5r/n units/source, plus thin work and transport: 1.875 at r=384, 1.094 at r=224. It still does not make old content free.

Loss: none if the stated identities hold. This differs from dead item 11: it combines equal right factors in slice formulas; it never constructs a symmetric tensor. **First falsifier:** inspect the actual symbolic M formula and operation trace (0 units). If already fused, there is no saving. Otherwise verify one cached source against the original contraction, under 7 units. No claim is made that V29 lacks this optimization.

Q2. Three ranked, falsifiable explanations

The 133–165-unit interval does not establish a free old tier: 153 is an implementation subtotal, not a lower bound. The first two hypotheses carry the old information unchanged.

1. Precision accounts for much of the bill (conjecture). If fraction f of V29's *metered cost* is float64 work that can safely become float32, cost becomes 260(1−f/2). Reaching 133–165 requires f=0.73–0.98; all-float64 conversion gives 130 units. **First falsifier:** a dtype census, 0 units. If this condition passes, one controlled rerun costs at most 260 units; require final-mean RMS drift below 10^{-6}. Signature: near-uniform per-layer cost reduction, including old sources. Better raw MSE still requires a separate closure improvement.

2. Exact contraction reuse (conjecture). Q1(d)'s fusion preserves the source state. A 50% reduction in eligible work alone requires 190–254 eligible units to save 95–127 units. **First falsifier:** audit that subtotal and existing fusion, 0 units. A smaller subtotal rejects fusion as the sole explanation. Signature: identical means up to rounding, but cost still grows with old-source count.

3. Joint bias compensation (weak conjecture). A short-history chain might retune $\lambda_\ell \to a\lambda_\ell$ inside fourth-cumulant regeneration. Cost stays about 153 units; the old state is absent, with its effect only compensated in the mean. This escapes the window result *only if* regeneration was held fixed. If jointly retuned there, it is already dead; output-feature corrections alone do not test this internal response. Nevertheless, reducing 10^{-6} to 1.9×10^{-8} requires explaining 98.1% of error energy.

First falsifier: run a=0,1,2, costing 459 units. Using the existing ground truth only as an oracle diagnostic, project the baseline error onto the two output-response differences. If even that span misses the target, reject the local-response version. This is not a deployable ground-truth-trained correction. Signature: a narrow joint optimum and substantial D21 error despite low final-mean error. I rank this last; it is not evidence that the dead window method works.

Q3. The single measurement I would make next

Measure the conditional variance of contracted old content, not another singular spectrum of its products. Start with one age≥7 source at layer 10. This directly tests whether the apparent product "noise" is negligible after contraction or is the missing quenched signal.

Freeze that checkpoint's gates and source legs. Apply the gates, and obtain τ and $\|T\|^2$ for the resulting source tensor. They can be computed from the six pairwise leg Grams A^TA, P^TP, M^TM, A^TP, A^TM, P^TM, plus elementwise contractions. This avoids materializing n^3 entries. The trace follows by contracting two legs of each symmetrized source term.

Resample 32 independent blocks of fresh weight rows, using 64 rows for the repeated index i and a disjoint 64 for c. Compute each 64×64 exact source D21 block, and its conditional prediction $E_g[D21|h]$. Average predictions conditionally on h, or use the analytic trace prediction directly; do not compare a nonzero fixed-network tensor against an unconditional ensemble mean of zero.

On the actual next-layer weights, also record the prediction residual normalized by *total* D21 RMS. Record D3 separately. Report three numbers: the analytic ρ from Q1(b), its resampling estimate with block-level uncertainty, and the actual-network residual/total-D21 ratio. A ratio above 0.02 immediately rejects this expectation-only replacement for that contribution; a smaller ratio does not establish that all old sources jointly pass.

FLOP budget. Six dense Grams cost about 6 units. Transporting three legs through 32 blocks of 128 rows costs 12 units. Four 64×64 slice contractions per block cost 0.5 units total. Allow about 7 units for the actual next-layer transport and slice evaluation, plus elementwise/thin overhead: **under 30 additional units**, or 6.45×10^{10} FLOPs. Reuse your saved checkpoint; if unavailable, add the measured 260-unit V29 run. At a sustained 1 GFLOP/s, that is approximately 65 seconds extra, or 10.4 minutes including the baseline.

Decision. If measured residual energy matches the derived variance and exceeds the total-slice tolerance, stop pursuing expectation-only self-averaging. The outstanding problem is preserving realized mixed cubic information. If it is unexpectedly small, first locate the conditioning or cancellation responsible; that is the specific new mechanism missing from the dead closures. Neither a flat product spectrum nor covariance alignment of a fourth-order core supplies it.

No proposed representation here has a complete, credible path to the leaders' cost and accuracy. The streamed identity and conditional fusion are implementation checks; the variance audit is the highest-information research experiment.

[1] Gaussian pairing identity: Konstantinos Mamis, "New formulas for moments of the multivariate normal distribution extending Stein's lemma and Isserlis theorem," §2.2, arXiv:2202.00189. The conditional-variance application and bounds above are derived here; this reference is not a claimed solution to quenched network cumulant transport.